

Day 116 — TNPSC Group 2A Study Material

Part A – General Studies: Current Affairs in History & National Movement (11): Vijayanagara-Era Inscriptions Found at Tirupati

READING MATERIAL

Covers the discovery of 2 Vijayanagara-era inscriptions in Tirupati, revealing 16th-century temple administration practices.

Q: Where were the 2 Vijayanagara-era inscriptions discovered, and when was this announced?

A: In the Seshachalam forest region of Tirupati district — announced 27 June 2026, found during a heritage exploration and documentation exercise by researchers and epigraphists.

Q: Which Vijayanagara Emperor is named in the inscriptions, and what do they document about his reign?

A: Emperor Sadasiva Deva Raya (mid-16th century) — the inscriptions document land grants issued during his reign for the maintenance of religious activities at the Parashurameswara Mandir in Gudimallam village.

Q: What is significant about the Parashurameswara Temple at Gudimallam itself?

A: It is renowned for housing one of the OLDEST known Shiva lingas in India — the new inscriptions show the temple continued receiving state patronage centuries after its original foundation, well into the Vijayanagara period.

Q: What do the inscriptions reveal about Vijayanagara governance structures?

A: That land grants required careful documentation, surveying, and administrative oversight — illustrating how imperial authority extended into local communities through networks of officials, village institutions, and temple (Mandir) administrations.

Q: What was the practical purpose of the land grants described in the inscriptions?

A: They provided income through agricultural production to support the temple's daily rituals, offerings, and maintenance.

MOCK TEST (WITH ANSWERS)

1. The 2 newly discovered Vijayanagara-era inscriptions were found in which forest region?

A. Seshachalam forest, Tirupati ✓

B. Mudumalai forest

C. Anamalai forest

D. Kalakad forest

Explanation: Seshachalam forest, Tirupati district. [medium, web-research]

2. Which Vijayanagara Emperor is named in these inscriptions?

A. Krishnadeva Raya

B. Achyuta Deva Raya

C. Sadasiva Deva Raya ✓

D. Venkata II

Explanation: Sadasiva Deva Raya. [medium, web-research]

3. The inscriptions document land grants for which temple?

A. Tirumala Venkateswara Temple

B. Parashurameswara Mandir, Gudimallam ✓

C. Brihadeeswarar Temple

D. Meenakshi Amman Temple

Explanation: Parashurameswara Mandir, Gudimallam. [medium, web-research]

4. The Parashurameswara Temple at Gudimallam is renowned for housing:

A. India's largest Nandi statue

B. One of the oldest known Shiva lingas in India ✓

C. The first Vijayanagara-built gopuram

D. A unique solar calendar inscription

Explanation: One of the oldest known Shiva lingas in India. [medium, web-research]

5. What practical purpose did the land grants described in the inscriptions serve?

A. Funding military campaigns

B. Providing agricultural income to support temple rituals, offerings, and maintenance ✓

C. Paying off war debts

D. Building new forts

Explanation: Providing agricultural income for temple rituals, offerings, and maintenance. [medium, web-research]

Part B – Aptitude & Mental Ability: Lowest Common Multiple (LCM)

READING MATERIAL

LCM is 1 of 10 named Aptitude sub-skills. Real exam evidence shows two distinct shapes: algebraic LCM-from-a-clue problems, and 'remainder' word problems — both are taught below as separate methods.

Q: Real exam (2025): If the LCM of 'a' and 'a+2' is 1300, find the LCM of 'a' and 'a+1'.

A: Since a and a+2 differ by 2, their HCF is either 1 or 2. If $\text{HCF}=2$: $\text{LCM} = a(a+2)/2 = 1300 \rightarrow a(a+2) = 2600$. Solving $a^2+2a-2600=0$ gives $a=50$ (check: $50 \times 52/2 = 1300$ ✓). Now find $\text{LCM}(50, 51)$: consecutive integers are always coprime ($\text{HCF}=1$), so $\text{LCM} = 50 \times 51 = 2550$.

Q: Method — 'least number leaving given remainders' (remainders differ from the divisors by a constant): Find the least number which, when divided by 16, 18, and 21, leaves remainders 3, 5, and 8 respectively.

A: Check the gap between each divisor and its remainder: $16-3=13$, $18-5=13$, $21-8=13$ — all the same gap! When that happens, the answer is $\text{LCM}(16,18,21) - 13$. $\text{LCM}(16,18,21) = 1008$. Answer = $1008-13 = 995$.

Q: Method — 'least number leaving the SAME remainder in every case': Find the least number which, when divided by 6, 7, 8, 9, and 12, leaves the same remainder 1 in every case.

A: When the remainder is identical across all divisors, the answer is $\text{LCM}(\text{divisors}) + \text{remainder}$. $\text{LCM}(6,7,8,9,12) = 504$. Answer = $504 + 1 = 505$.

Q: Method — 'mixed condition' (a common remainder for some divisors, but exact division for another): Find the least number which, when divided by 5, 6, 7, and 8, leaves remainder 3 in each case, but leaves NO remainder when divided by 9.

A: First find the least number leaving remainder 3 for 5,6,7,8: that's $\text{LCM}(5,6,7,8)+3 = 840+3 = 843$. But we also need it divisible by 9. Keep adding $\text{LCM}(5,6,7,8)=840$ repeatedly to 843 until the result is divisible by 9: 843 (no), 1683 ($1683 \div 9 = 187$, yes!). Answer = 1683.

Q: Method — LCM of numbers given as prime factorizations: Find the LCM of $(2 \times 3 \times 5 \times 7)$ and $(3 \times 5 \times 7 \times 11)$.

A: LCM takes the HIGHEST power of every prime that appears in either number. Primes involved: 2,3,5,7,11 — each appears at most once in either factorization, so $\text{LCM} = 2 \times 3 \times 5 \times 7 \times 11$ (just the union of all prime factors, since none repeat).

MOCK TEST (WITH ANSWERS)

1. If the LCM of 'b' and 'b+2' is 180, and their HCF is 2, find the LCM of 'b' and 'b+1'.

A. 342 ✓

B. 380

C. 361

D. 324

Explanation: $\text{LCM} = b(b+2)/\text{HCF} = 180 \rightarrow b(b+2) = 360 \rightarrow b^2+2b-360=0 \rightarrow b=18$ (check: $18 \times 20/2 = 180$ ✓). Now $\text{LCM}(18,19)$: consecutive integers are coprime, so $\text{LCM} = 18 \times 19 = 342$. [hard, authored]

2. Find the least number which, when divided by 12, 15, and 20, leaves remainders 7, 10, and 15 respectively.

A. 55 ✓

B. 60

C. 65

D. 53

Explanation: Gaps: $12-7=5$, $15-10=5$, $20-15=5$ — all equal, so answer = $\text{LCM}(12,15,20) - 5 = 60 - 5 = 55$. [medium, authored]

3. Find the least number which, when divided by 4, 5, and 6, leaves the same remainder 2 in each case.

A. 58

B. 60

C. 62 ✓

D. 64

Explanation: $\text{LCM}(4,5,6) = 60$. Answer = $60 + 2 = 62$. [easy, authored]

4. Find the LCM of $(2 \times 2 \times 3 \times 5)$ and $(2 \times 3 \times 3 \times 7)$.

A. $2 \times 2 \times 3 \times 3 \times 5 \times 7$ ✓

B. $2 \times 3 \times 5 \times 7$

C. $2 \times 2 \times 3 \times 5 \times 7$

D. $4 \times 9 \times 5 \times 7$

Explanation: Take the highest power of each prime: 2^2 (from the first), 3^2 (from the second), 5^1 , 7^1 . $\text{LCM} = 2^2 \times 3^2 \times 5 \times 7$, which equals option (A) written out as $2 \times 2 \times 3 \times 3 \times 5 \times 7$. [medium, authored]

5. The LCM of two consecutive integers n and $n+1$ is always:

A. $n + 1$

B. $n(n+1)$ ✓

C. $n - 1$

D. Cannot be determined

Explanation: Consecutive integers are always coprime ($\text{HCF}=1$), so their LCM is simply their product, $n(n+1)$. [easy, authored]

Part C – General English: Prose: "Tech Bloomers"

READING MATERIAL

A non-fiction piece about how assistive technology empowers people with disabilities, illustrated through two individuals: Alisha, who uses voice-dictation software to study independently, and David, who has Athetoid Cerebral Palsy and uses eye-movement-controlled devices to communicate, study, and even play adaptive sports.

Q: What is the central theme of "Tech Bloomers"?

A: The use of technology for empowerment — specifically, how assistive technology enables people with disabilities to study, communicate, and participate fully in life.

Q: What has technology helped Alisha do, and what software does she use?

A: Technology has helped Alisha study independently. She uses 'Dragon Dictate', a voice-dictation software.

Q: Who introduced Dragon Dictate to Alisha, what is that person's role, and how long did Alisha take to train it?

A: Kim, an Assistive Technologist, introduced Dragon Dictate to Alisha. Alisha took just a few hours to train the software to her voice.

Q: What condition was David born with, and how does he control his 'Liberator' device?

A: David was born with Athetoid Cerebral Palsy. He controls his Liberator device using eye movements.

Q: What additional controller is found in David's chair, and what system helps him select icons with his eyes?

A: A Headrest ACTIV controller is found in his chair. The 'ECO2' system, using an 'ECO point', helps him select icons with his eyes.

Q: What does David use ECO2 for in daily life, and which sport does he NOT play (among the adaptive sports mentioned)?

A: He uses ECO2 to access his curriculum (study materials) and to order food. Among the sports mentioned (Boccia, Hockey, Tennis), he does NOT play Tennis — he plays Boccia and Hockey.

MOCK TEST (WITH ANSWERS)

1. The central theme of "Tech Bloomers" is:

- A. History of technology
- B. Role of AI in warfare
- C. Use of technology for empowerment ✓**
- D. Business automation

Explanation: Use of technology for empowerment. [easy, web-research]

2. Technology has helped Alisha to:

- A. Cook food
- B. Type with her hands
- C. Speak multiple languages
- D. Study independently ✓**

Explanation: Study independently. [medium, web-research]

3. The voice-dictation software Alisha uses is called:

- A. SpeakEasy
- B. Dragon Dictate ✓**
- C. VoiceWrite
- D. Dictation Pro

Explanation: Dragon Dictate. [medium, web-research]

4. Who introduced Dragon Dictate to Alisha, and what was that person's role?

- A. Her doctor
- B. Her classmate
- C. Kim, an Assistive Technologist ✓**
- D. David, an author

Explanation: Kim, an Assistive Technologist. [medium, web-research]

5. How long did it take Alisha to train Dragon Dictate to her voice?

- A. A week
- B. A few hours ✓**
- C. A month
- D. A day

Explanation: A few hours. [medium, web-research]

6. David, in "Tech Bloomers", was born with which condition?

- A. ADHD
- B. Spina bifida
- C. Athetoid Cerebral Palsy ✓**
- D. Dyslexia

Explanation: Athetoid Cerebral Palsy. [medium, web-research]

7. David controls his 'Liberator' device using:

- A. Voice
- B. Hand movements
- C. Eye movements ✓**
- D. Foot taps

Explanation: Eye movements. [medium, web-research]

8. What additional controller is found in David's chair?

- A. Wrist band
- B. Headrest ACTIV controller ✓**
- C. Palm switch
- D. Shoulder joystick

Explanation: Headrest ACTIV controller. [hard, web-research]

9. Which system helps David select icons using his eyes, and what does he use it for?

- A. Dragon Eye — for gaming
- B. SmartSight — for navigation
- C. ECO2 with ECO point — to access curriculum and order food ✓**
- D. BlinkAssist — for typing only

Explanation: ECO2 with ECO point, used to access his curriculum and order food. [hard, web-research]

10. Which sport does David NOT play, among those mentioned in the passage?

- A. Boccia
- B. Hockey
- C. Tennis ✓**
- D. All of the above are played

Explanation: Tennis — he plays Boccia and Hockey. [medium, web-research]